Capital Budgeting
& Project Financing

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WAKE FOREST
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Key Phrase

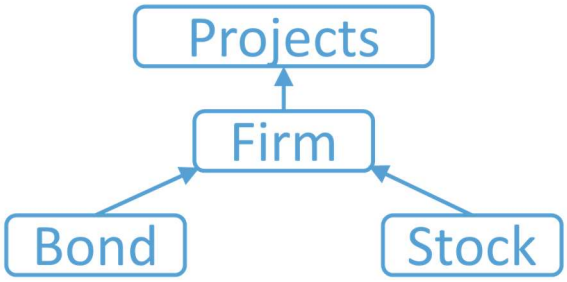
Value any asset by discounting its expected future cash flows back at an appropriate, risk adjusted discounted rate

$$Price = PV(Future\ Cash\ Flows)$$
$$= \sum_{t=0}^{\infty} \frac{CF_t}{(1+r)^t}$$

- Two ingredients
  - Cash flows
  - Discount rate

Course Objectives

- How do we price assets?
  - Time Value of Money
  - Bond & Stock Valuation
- How do we optimally raise and invest capital?
  - Project Financing
  - Capital Budgeting



Investment Rules

- Decision: Should the firm invest in this project?
  - Does the project create value for shareholders?
- Conclusion: Accept all and only projects with a positive NPV

Decision Rules

- Payback Period
  - Number of periods it takes to recover initial investment
  - Accept if payback period is less than subjective threshold
- NPV
  - $NPV = PV(CF_t)$
  - Accept if $NPV > 0$
- IRR
  - r such that $NPV = 0$
  - Accept if IRR $>$ cost of capital
- Hurdle Rate
  - r^{Hurdle} adds a project risk-premium to the cost of capital
  - Accept if $\text{IRR} > r^{Hurdle}$

Decision Rules: Example

Project costs \$100 today and pays \$50 for next three years. $r = 10\%$

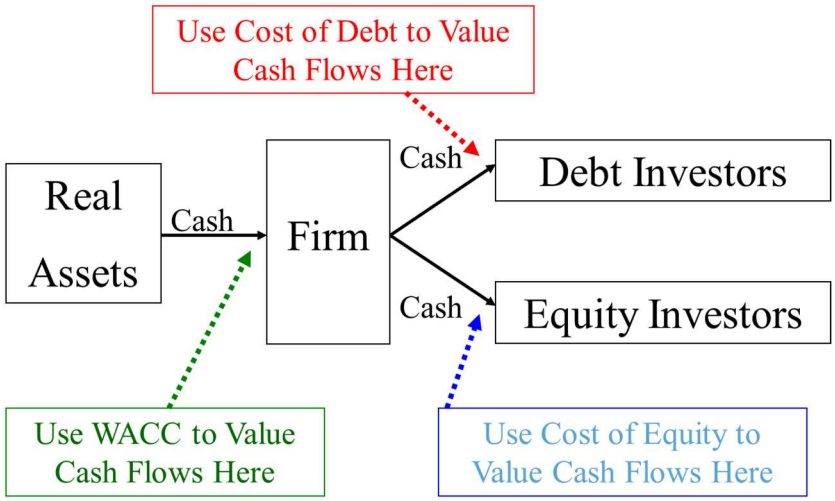
- Payback Period $= 2 \Rightarrow \$0 = -\$100 + \$50 + \$50$
- $NPV = -\$100 + \frac{\$50}{(1+10\%)^1} + \frac{\$50}{(1+10\%)^2} + \frac{\$50}{(1+10\%)^3} = \$24.34$
- $IRR = 23\% \Rightarrow \$0 = -\$100 + \frac{\$50}{(1+23\%)^1} + \frac{\$50}{(1+23\%)^2} + \frac{\$50}{(1+23\%)^3}$
- Hurdle Rate: $r^{Hurdle} = 10\% + rp$

Investment Rules

- NPV
  - Considers both time and risk of cash flows
  - Main rule we should use

- Payback Period
  - Subjective and does not consider time or risk
- Hurdle Rate
  - Subjective and rejects profitable projects
- IRR
  - Not good for complex cash flows
  - Not good for ranking mutually exclusive projects

Cost of Capital



Why Do We Care?

- If you overestimate the WACC, you will…
  - Reject profitable projects

- If you underestimate the WACC, you will…
  - Accept unprofitable projects

WACC - Weighted Average Cost of Capital

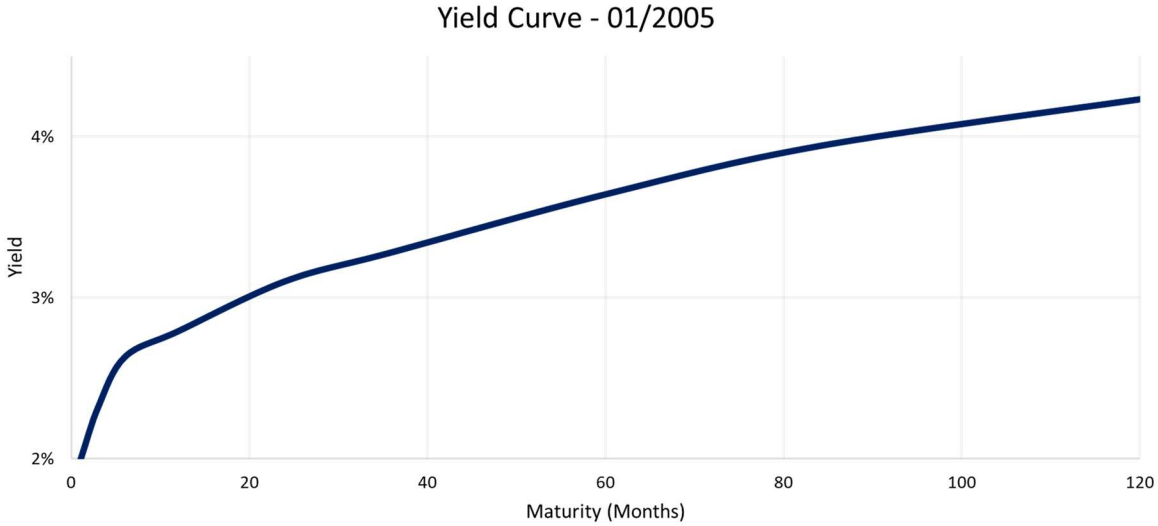
$$R_{WACC} = \frac{D}{D+E} R_D (1-t) + \frac{E}{D+E} R_E$$

- WACC: R_{WACC}
- Cost of debt: R_D
- Cost of equity: R_E
- Firm's tax rate: t
- Market value of debt: D
- Market value of equity: E

Cost of Debt R_D

$$R_D = R_f + Default\ Spread$$

Risk-Free R_f: U.S. Treasury Yield Curve

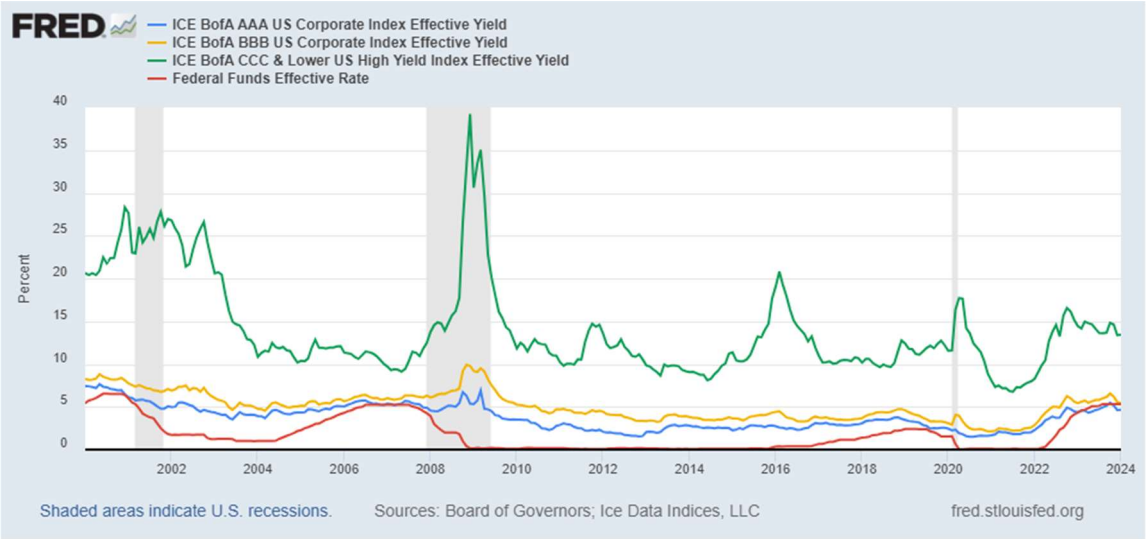


Default Spread: Bond Ratings

	Investment-Quality Bond Ratings				Low-Quality, Speculative, and/or "Junk" Bond Ratings					
	High Grade		Medium Grade		Low Grade		Very Low Grade			
Standard & Poor's	AAA	AA	A	BBB	BB	B	CCC	CC	C	D
Moody's	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C	

Moody's	S&P	
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in higher-rated categories.
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.
Ba; B Caa Ca C	BB; B CCC CC C	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and Ca, CC, and C the highest degree of speculation. Although such debt is likely to have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Issues rated C by Moody's are typically in default.
	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.

Credit Spreads over Time



Cost of Equity R_E

$$R_E = R_f + \beta_E \times \left(R_M - R_f\right)$$

- Risk-free Rate: R_f
- Market Risk Premium: $R_M - R_f$
- Equity beta: β_E

Financial Leverage

- Financial leverage (debt) magnifies the risk of a project for the shareholders

- The "asset beta" is an unlevered beta β_A
  - Measurement of the operational risk

- The "equity beta" is a levered beta β_E
  - Measurement of the total risk, including the impact of financial (leverage) risk
  - Financial leverage magnifies the risk of a project for the shareholders

- The relationship

$$\beta_E = \beta_A \left[1 + \frac{D}{E} \times (1 - t)\right]$$

Capital Structure & Financial Leverage

In practice, there are two common times a manager needs to account for financial leverage in their cost of equity calculations:

- When you are changing your capital structure, or
  - Need to unlever and relever based on your old and new capital structure
- When you are investing in a new business model whose risk profile is different from your existing line of business.
  - Need to identify a pure play, measure their equity beta, and unlever with their capital structure
  - Relever with your own capital structure to find your own equity beta

Optimal Capital Structure

- Increased use of debt has the following results:
  - Higher financial leverage will increase the value of debt's tax shield
  - Higher financial leverage may increase cost of debt due the increase likelihood of bankruptcy
  - In practice, mangers should choose their capital structure such that they equate the marginal benefit of debt's tax shield with the marginal cost of bankruptcy

Summary

- WACC

$$R_{WACC} = \frac{D}{D+E} R_D (1-t) + \frac{E}{D+E} R_E$$

- Cost of Debt: $R_D = R_f + RP_D$
- Cost of Equity: $R_E = R_f + \beta_E \times RP_M$

- Equity Beta

$$\beta_E = \beta_A \left[1 + \frac{D}{E} \times (1-t) \right]$$